

Emerging Technologies for the Circular Economy

Lecture 3: Lifecycle Assessment (LCA) - Part 1

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INTRODUCTION

LCA - Motivation



LCA - Motivation



**Battery Electric Vehicles
(EV)**

Or

**Internal Combustion Engine
Vehicles**

EV Break-Even Point?

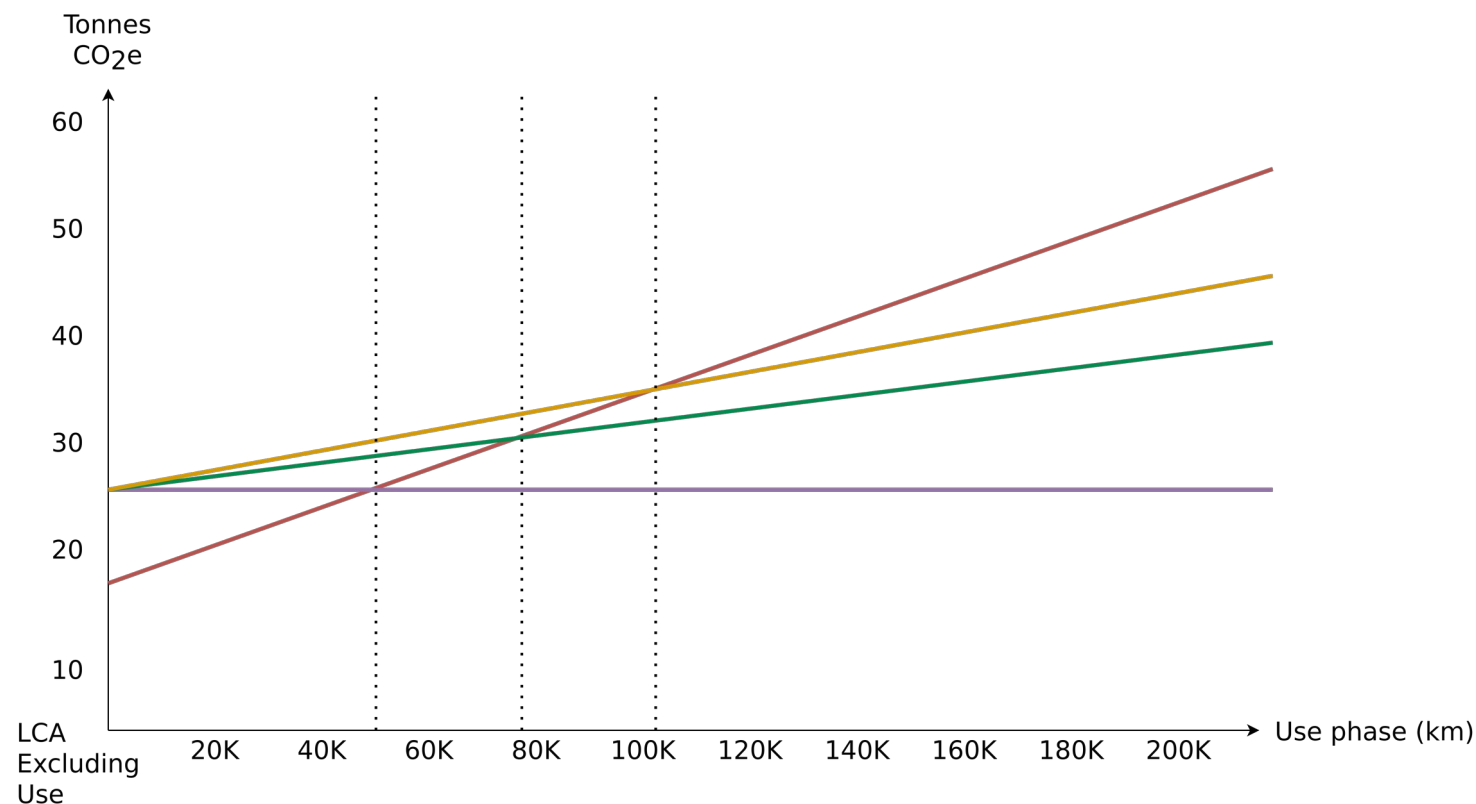
What is the **break-even** point (in km) after which an EV would have caused fewer emissions than an Internal Combustion Engine (ICE?)

- a. 0 – 50.000km
- b. 50.000 – 100.000km
- c. 100.000 – 150.000km
- d. 150.000 – 200.000km
- e. After 200.000km

Life Cycle Assessment - Polestar 2

Cumulative amount of GHGs emitted depending on total km driven, from Polestar 2 (with different electricity mixes)

- XC40 ICE
- Polestar 2 -- Global electricity Mix
- Polestar 2 -- European (EU28) electricity Mix
- Polestar 2 -- Wind Power



Number of kilometers driven at break-even between Polestar 2 with different electricity mixes in the use phase of XC40 ICE (petrol)	Electric mix	Break-even (km)
	Polestar 2 -- Global electricity Mix	112,000
	Polestar 2 -- European (EU28) electricity Mix	78,000
	Polestar 2 -- Wind Power	50,000

LIFECYCLE ASSESSMENT (LCA)

Lifecycle Assessment (LCA)

Definition

“LCA addresses the environmental aspects and potential environmental impacts (e.g. use of resources and environmental consequences of releases) throughout a product’s lifecycle from raw material acquisition through production, use, end-of-life treatment, recycling and final disposal (i.e., cradle-to-grave).” -- ISO 14040



Lifecycle Assessment (LCA)

ISO 14040 & ISO 14044

- The ISO 14040/14044 (ISO14040, 2006) (ISO14044, 2006) together provide a *loose* methodology for conducting LCA studies.
- ISO 14040 defines the *principles and framework* of the standard
- ISO 14044 provides *requirements and guidelines* for LCA practitioners.

- Their scope is very broad, hence requiring LCA practitioners to further refine the methodology for their specific needs.

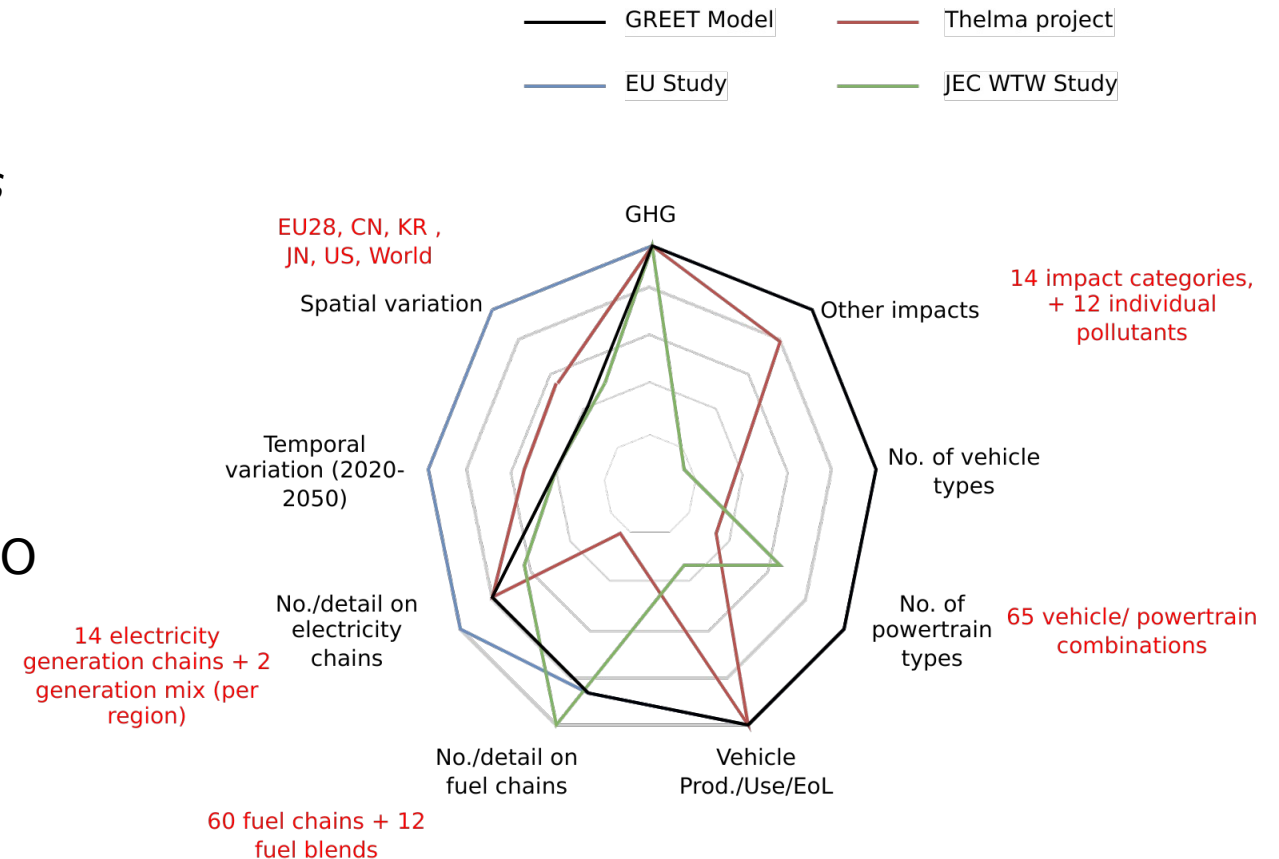
ISO 14040 Environmental management – Life cycle assessment – Principles and framework, International standards organisation (<https://www.iso.org/standard/37456.html>)

ISO 14044 Environmental management – Life cycle assessment – Requirements and guidelines, International standards organisation (<https://www.iso.org/standard/38498.html>)

Lifecycle Assessment (LCA)

2020 EU Commission Report

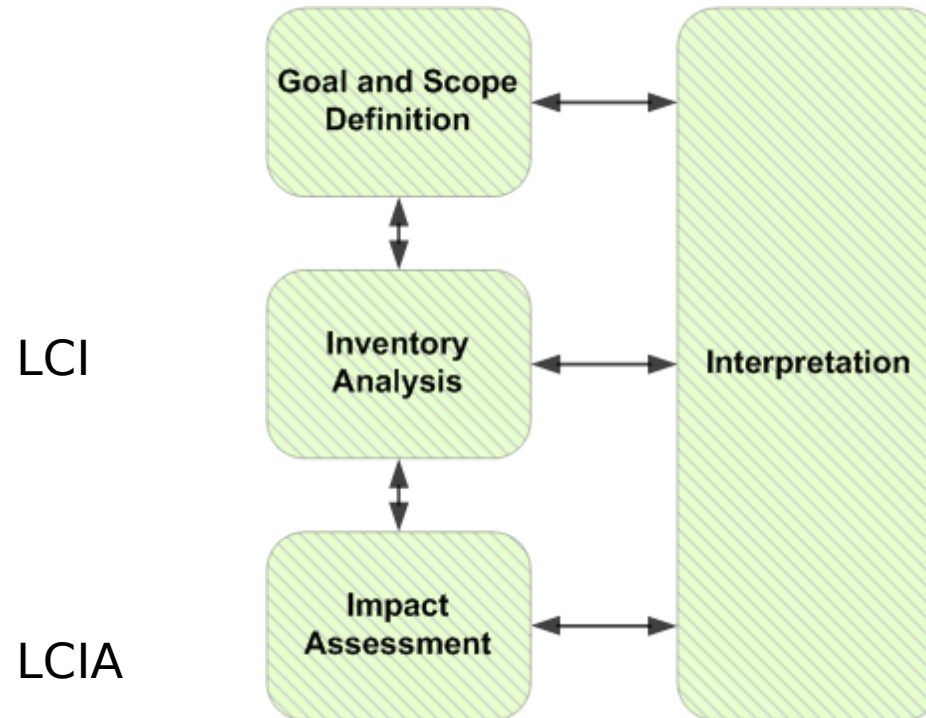
- “Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA”
 - Prepared by *Ricardo Energy and Environment*
 - Generally follows the ISO 14040 and ISO 14044 standards.



Overview of the LCA Application framework and key data flows, in comparison with other studies

Lifecycle Assessment (LCA)

The four main stages



GOAL AND SCOPE DEFINITION

Goal and Scope Definition

Goal of an LCA study

- **ISO 14040 definition**
- The goal of an LCA states:
 - The intended application
 - The reasons for carrying out the study
 - The intended audience
 - Whether the results are intended to be used in comparative assertions released publicly

Goal and Scope Definition

Goal of an LCA study

- **2020 EU Study:**

- The intended application: A representative selection of road vehicle configurations.
- Aims to enhance the Commission's understanding of environmental impacts and of suitable methodologies to assess them in the mid- to long-term time frame (until 2050).
- Target audience: European Commission and decision-makers.

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Scope of an LCA study

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quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

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quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

measure of the outputs from processes in a given product system required to deliver the performance defined by the functional unit

Goal and Scope Definition

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quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

measure of the product(s) or product parts required to deliver the performance defined by the functional unit.

Goal and Scope Definition

Scope of an LCA study

- Dishwasher



- Handwash



Vs.

Which is better?

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- **Dishwasher vs Handwash?**
- Functional units and reference flows
 - Functional unit (quantifiable performance of the system)?
 - Reference Flows?



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- **Dishwasher vs Handwash?**
- Functional units and reference flows
 - Functional Unit: 2 million dishes, cups, cutlery washed.
 - Reference Flows?



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- **Dishwasher vs Handwash?**
- Functional units and reference flows
 - Functional Unit: 2 million dishes, cups, cutlery washed.
 - Reference Flows: Water, heating, soap, electricity, (everything used to manufacture and maintain a dishwasher)



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- **2020 EU Study**
- Functional units and reference flows
 - Based on vehicle size and utility

Body Type	Passenger Car	Van	Rigid Lorry	Artic Lorry	Urban bus	Coach
Default reference flow	Vehicle-km (vkm)	Vehicle-km (vkm)	Tonne-km (tkm)	Tonne-km (tkm)	Vehicle-km (vkm)	Vehicle-km (vkm)

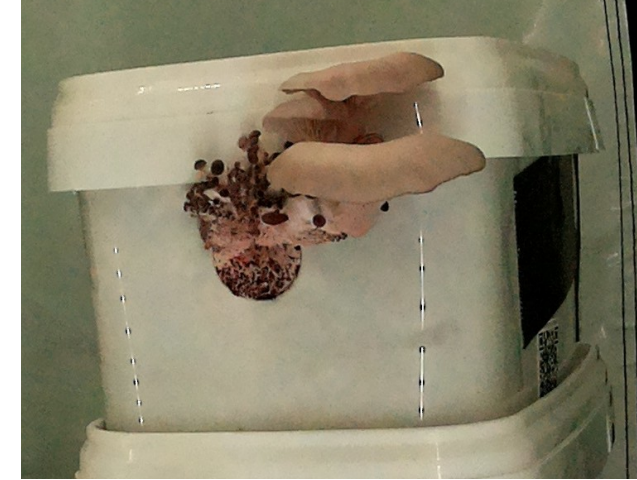
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MushR: A Short Detour

Mushroom production

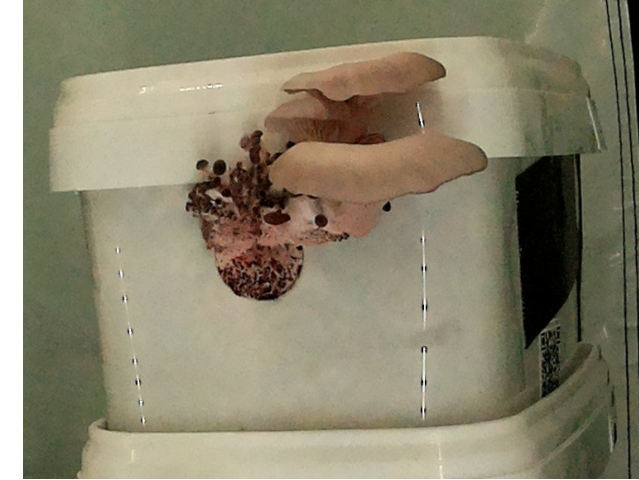
- **Substrate:**
 - The medium from which the mushrooms grow.



MushR: A Short Detour

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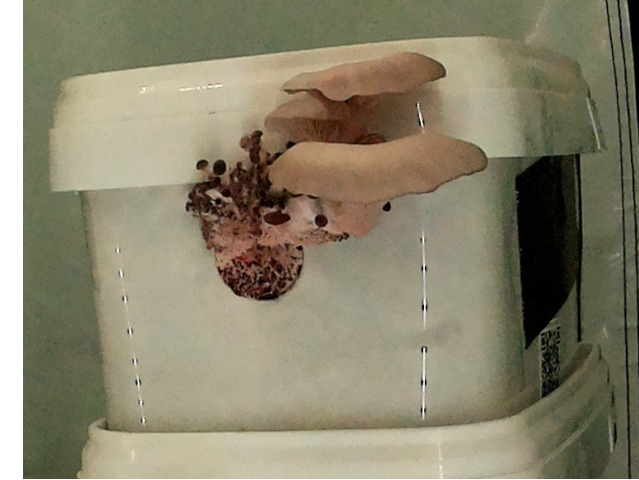
- **Substrate:**
 - The medium from which the mushrooms grow.
- **Substrate Container:**
 - Contains the substrate throughout the entirety of the substrate's "lifespan".
 - After inoculation:
 - Small ventilation holes (filtered) that facilitate incubation.



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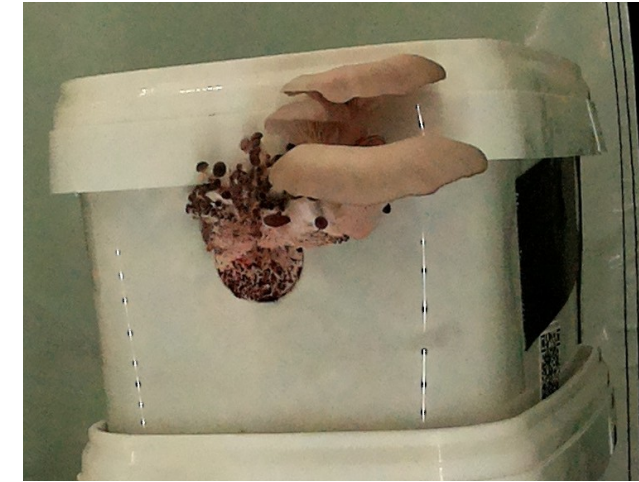
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- **Substrate Container:**
 - Contains the substrate throughout the entirety of the substrate's "lifespan".
 - After inoculation:
 - Small ventilation holes (filtered) that facilitate incubation.
 - After incubation period:
 - Fruiting holes are opened.
 - Contained substrate is exposed to fresh air and high humidity allowing mushrooms to grow through the holes.



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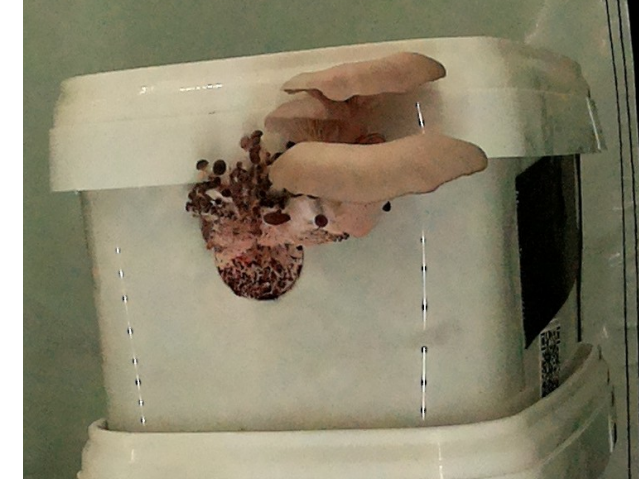
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 - After inoculation:
 - Small ventilation holes (filtered) that facilitate incubation.
 - After incubation period:
 - Fruiting holes are opened.
 - Contained substrate is exposed to fresh air and high humidity allowing mushrooms to grow through the holes.
 - After some fruiting cycles:
 - Substrate is discarded/composted.



MushR: A Short Detour

Mushroom Substrate Containers

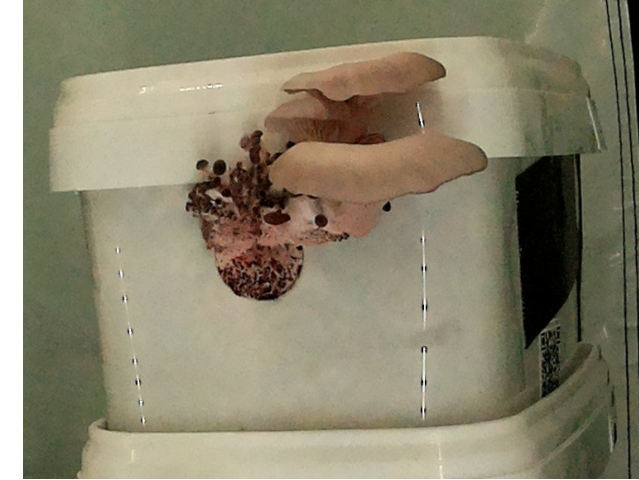
- **Traditional Substrate Containers:**
 - Plastic (polypropylene) bags
 - Usually sealed by folding the opening several times, then sealing it with tape/clips.
 - Can be tricky to seal.
 - Need to be re-opened for inoculation.



MushR: A Short Detour

Mushroom Substrate Containers

- **Traditional Substrate Containers:**
 - Plastic (polypropylene) bags
 - Usually sealed by folding the opening several times, then sealing it with tape/clips.
 - Can be tricky to seal.
 - Need to be re-opened for inoculation.
 - Fruiting holes are permanent; basically made by cutting through the bags → Cannot be reused
 - Single-use only.

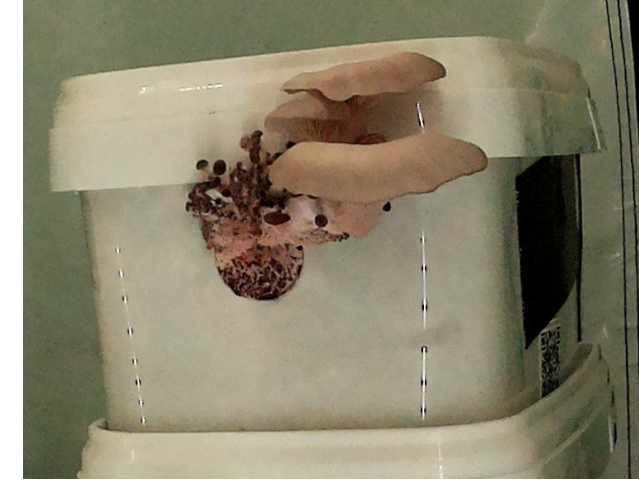


MushR: A Short Detour

Mushroom Substrate Containers

■ **MushR Substrate Pods:**

- Plastic (polypropylene) buckets.
- Sealed by a plastic lid.
 - Trivial to seal/unseal.
- Fruiting holes are still permanent; drilled into the bucket
 - Sealed with micro-porous tape during incubation.
 - Tape is removed for fruiting mushrooms → Can be reused.

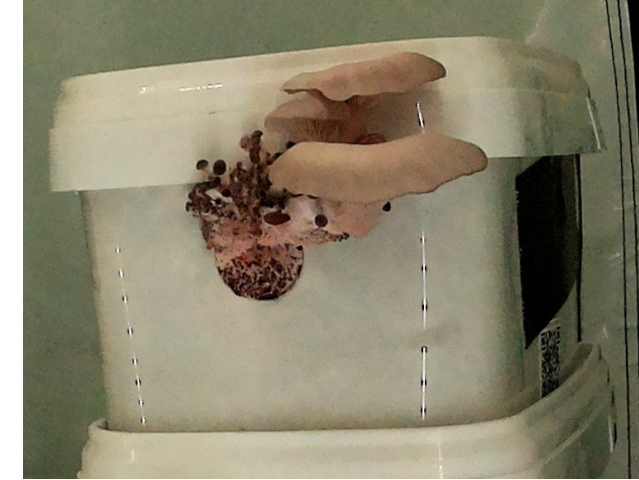


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Mushroom Substrate Containers

■ **MushR Substrate Pods:**

- Plastic (polypropylene) buckets.
- Sealed by a plastic lid.
 - Trivial to seal/unseal.
- Fruiting holes are still permanent; drilled into the bucket
 - Sealed with micro-porous tape during incubation.
 - Tape is removed for fruiting mushrooms → Can be reused.
- But:
 - Higher resource consumption required for production.
 - More complicated manufacturing process.

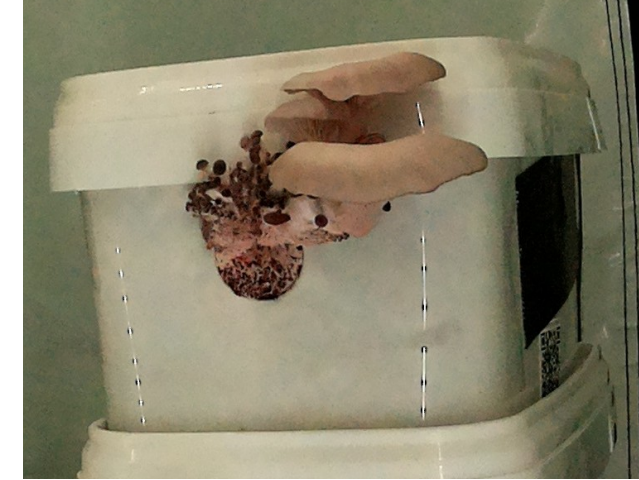


MushR: A Short Detour

LCA on Substrate Containers?

■ Goal

- Compare the environmental impact of the developed mushroom pods with non-reusable substrate bags by running lifecycle assessment calculations.



Goal and Scope Definition

Functional Unit VS Reference Flow

Functional Unit

quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

Reference Flow

a measure of the product(s) or product parts required to deliver the performance defined by the functional unit.

- How can you define the functional unit for a mushroom substrate container?

Goal and Scope Definition

Functional Unit VS Reference Flow

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- How can you define the functional unit for a mushroom substrate container?
 - In terms of “Colonizable volume of the container”: 3L colonizable volume

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- How can you define the functional unit for a mushroom substrate container?
 - In terms of “Colonizable volume of the container”: 3L colonizable volume
- How can you define the reference flow?

Goal and Scope Definition

Functional Unit VS Reference Flow

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quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

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- How can you define the functional unit for a mushroom substrate container?
 - In terms of “Colonizable volume of the container”: 3L colonizable volume
- How can you define the reference flow?
 - In terms of the “Amount of polypropylene (g)” (weight of the container)

Goal and Scope Definition

Scope of an LCA study

- **ISO 14040 definition**

- The scope of an LCA should describe:
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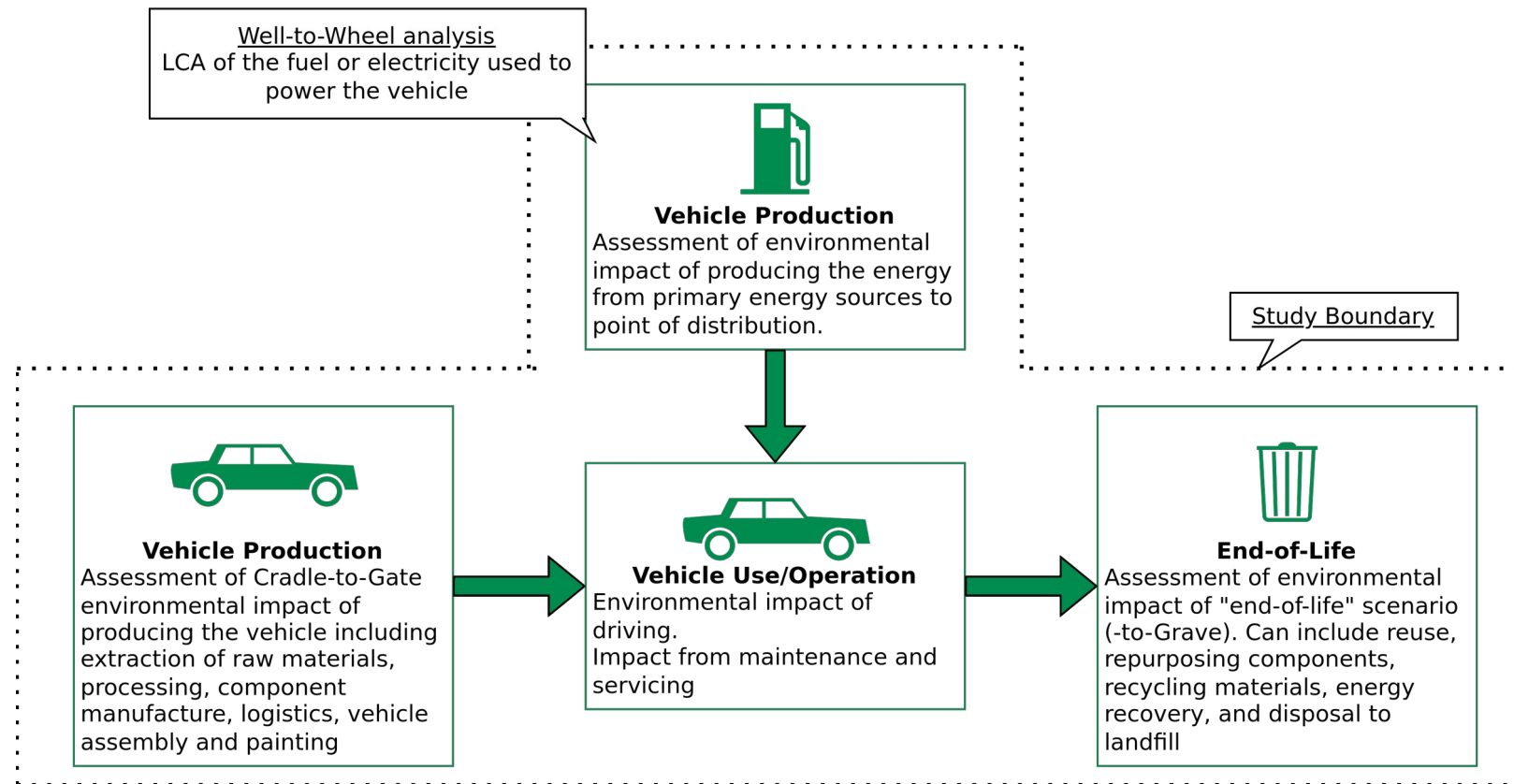
determines which processes
are included in the LCA in
accordance with it's goal

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Goal and Scope Definition

System Boundary of 2020 EU Study



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 - ...
- **MushR reusable mushroom pods**
 - (Theoretical) Manufacturing process
 - (Theoretical) End-of-life recycling process

Goal and Scope Definition

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▪ Dishwasher vs handwash?

- Commonalities
 - Water usage
 - Water distribution/treatment systems
 - Soap



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▪ Dishwasher vs handwash?

– Dishwasher

- Raw materials, manufacturing process, electricity consumption....
- Maintenance, Disposal
- Multiple models/manufacturers?



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▪ Dishwasher vs handwash?

- Handwash
 - Multiple handwash techniques?
 - Multiple people involved?
 - Multiple experiments?



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Goal and Scope Definition

Quantitative definition of system boundaries - the Cut-off criteria

- In general, all processes and flows that are attributable to the analysed system are to be included in the system boundaries.
- **However, not all of them are quantitatively “relevant”.**
- For less relevant ones, data of lower quality (estimates) can be used, limiting the effort for collecting high quality data.
- Irrelevant ones, can be entirely “Cut-off”



Goal and Scope Definition

Quantitative definition of system boundaries - the Cut-off criteria

- “Cut-off” refers to the omission of not relevant life cycle stages, activity types (e.g. investment goods, storage, ...), specific processes and products and *elementary flows* from the system model.
- Cut-offs are quantified in relation to the percentage of environmental impacts that is approximated to be excluded via the cut-off.
- e.g., "95 %" relates to cutting off about 5 % of the total environmental impact (or of a selected impact category)



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- Cut-offs are quantified in relation to the percentage of environmental impacts that is approximated to be excluded via the cut-off.
- e.g., “95 %” relates to cutting off about 5 % of the total environmental impact (or of a selected impact category)
- **BUT**, this would require an approximation of 100% of the impact, because if we already knew what 100% impact is, we wouldn’t be doing the study anyway.
- **IMPORTANT:** Cut-off should not be so big, or you can risk having incomplete data (meaning lower environmental impacts) and also overall uncertainty.

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▪ 2020 EU Study

Impact Category	Indicator and unit
Climate change	Greenhouse gas emissions GWP100 in CO ₂ eq (including carbon feedbacks)
Energy consumption	Cumulative energy demand in MJ: non-renewable (fossil and nuclear) and renewable
Acidification	Acidification potential in SO ₂ eq
Eutrophication	Eutrophication potential in PO ₄ ³⁻ eq
Photochemical ozone formation	Photochemical Ozone Creation Potential POCP in NMVOC eq
Ozone depletion	ODP in R11 eq
Ionising radiation	Ionising radiation potentials in U235 eq
Particulate matter	Particulate matter formation in PM2.5 eq
Human toxicity, cancer and non-cancer	Comparative Toxic Unit for Human Health in CTUh
Ecotoxicity, freshwater	Comparative Toxic Unit for ecosystems in CTUe
Resource depletion - minerals and metals	ADP ultimate reserves in Sb eq
Resource depletion - fossil energy carriers	ADP fossil in MJ
Land use	Land occupation in m ² * a
Water scarcity	Scarcity-adjusted water use in m ³

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 - ...
- **MushR reusable mushroom pods**
 - Limitations
 - No data calculations on the performance of the substrate container
 - Amount of mushrooms harvested?
 - Contamination rate?
 - Generic manufacturing and recycling data

Goal and Scope Definition

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- **Dishwasher vs handwash**

- Limitations
 - What if we don't have manufacturing and recycling data for Dishwashers?
 - What is the minimum number of people that we should do experiments with to get data for handwashing dishes?
 - We probably cannot count human effort?

Goal and Scope Definition

Scope of an LCA study

- **ISO 14040 definition**

- The scope of an LCA should describe:
 - The functional units of the system(s)
 - Reference flows
 - The system boundary
 - LCIA methodology and types of impacts analysed
 - **Limitations**
 - **Data quality requirements**
 - ...

- **Dishwasher vs handwash**

- Data Quality
 - We might only use estimates for manufacturing and recycling data, defining minimum quality requirements for the same.

LIFECYCLE INVENTORY ANALYSIS (LCI)

Lifecycle Inventory Analysis (LCI)

Definitions

LCI is the phase of lifecycle assessment involving the compilation and quantification of *inputs* and *outputs* for a product throughout it's lifecycle.

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Lifecycle Inventory Analysis (LCI)

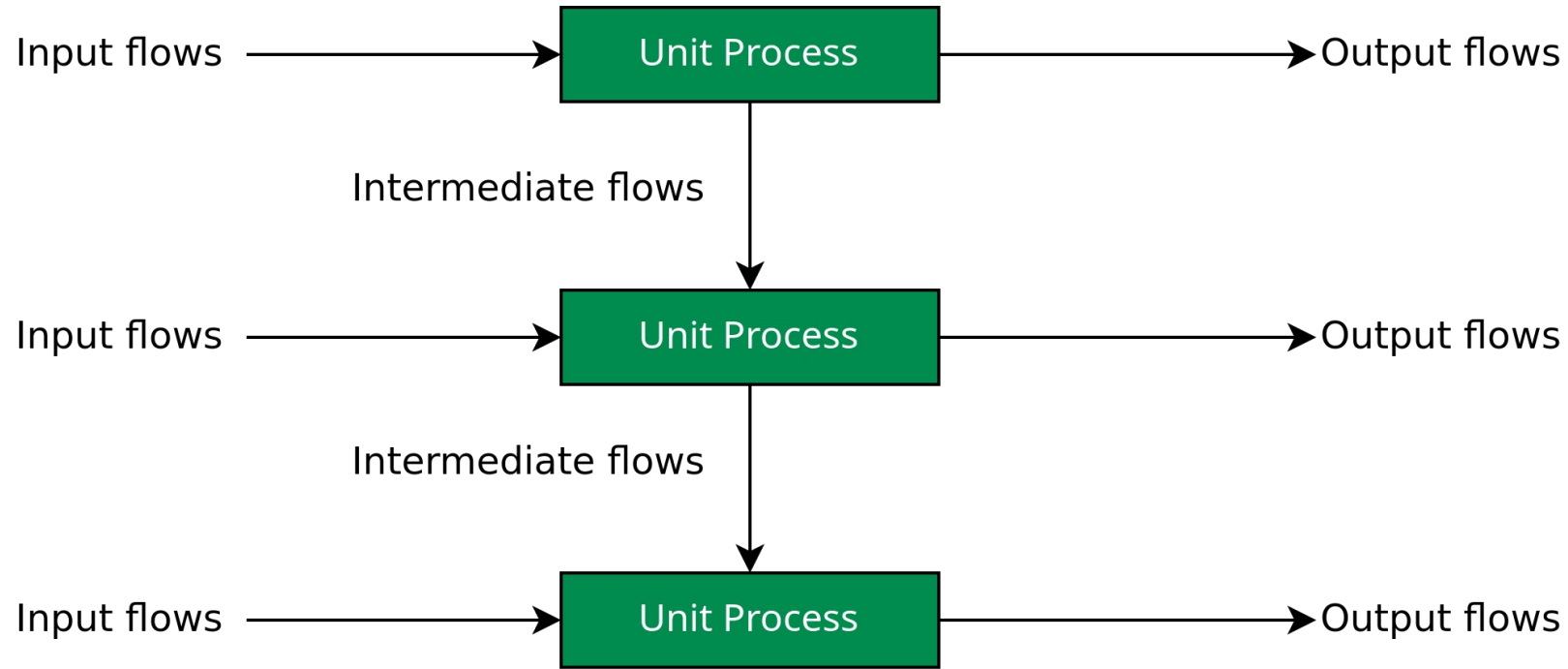
Definitions

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A *Unit Process* is the smallest element considered in the life-cycle inventory analysis for which input and output data are quantified.

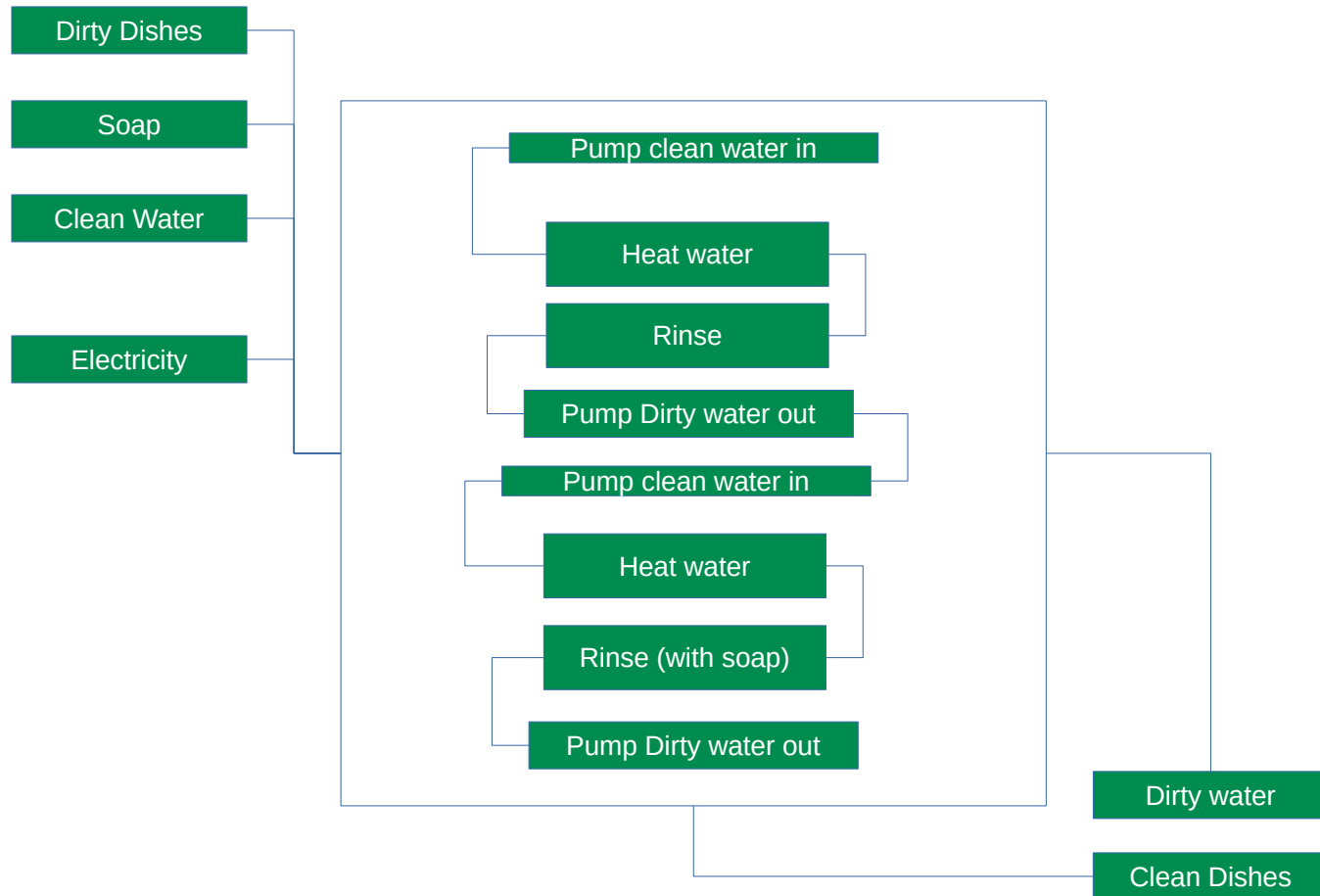
Lifecycle Inventory Analysis (LCI)



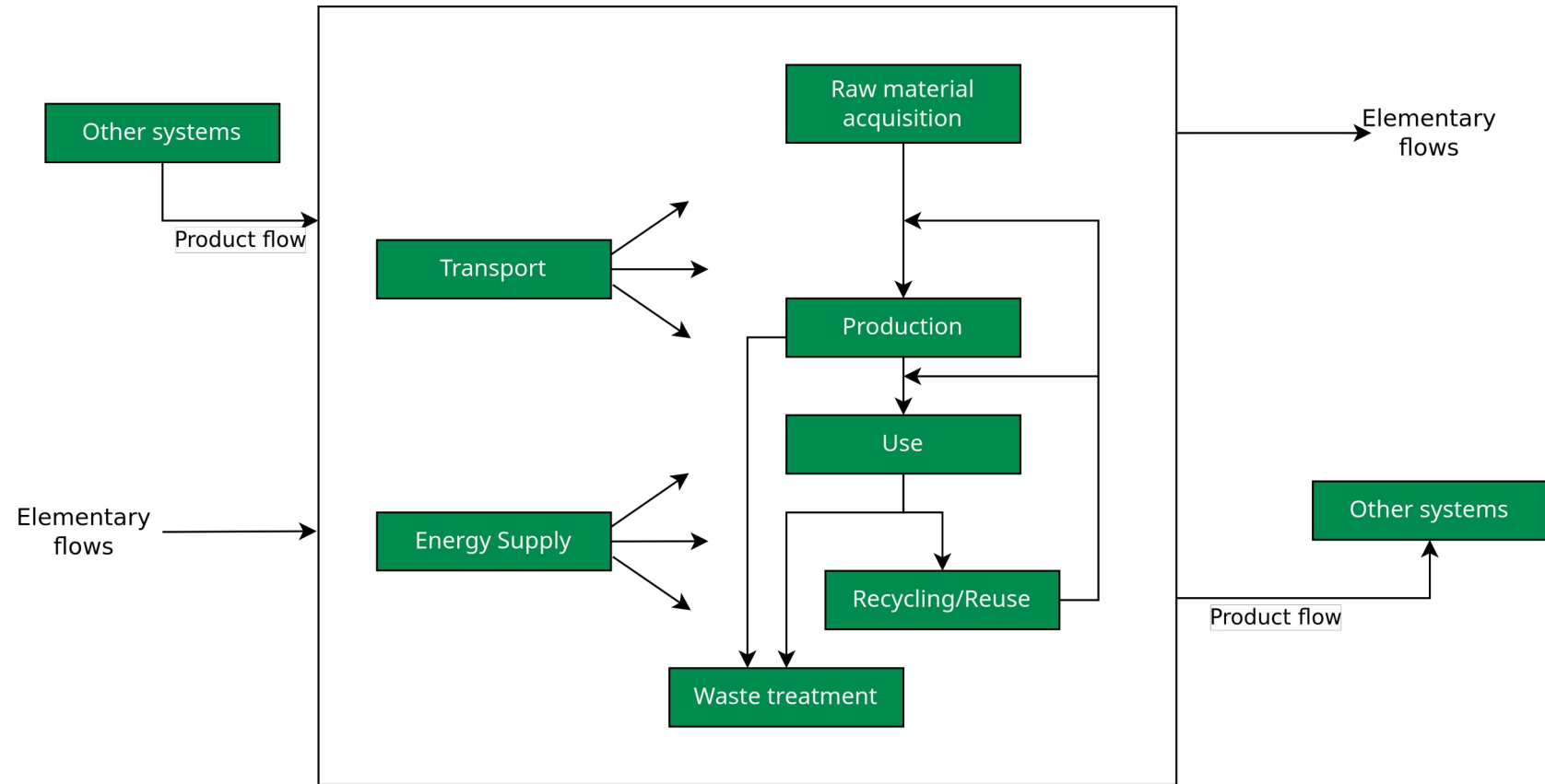
Example of a set of unit processes within a product system

Lifecycle Inventory Analysis (LCI)

Unit flows for a Dishwasher vs handwash?



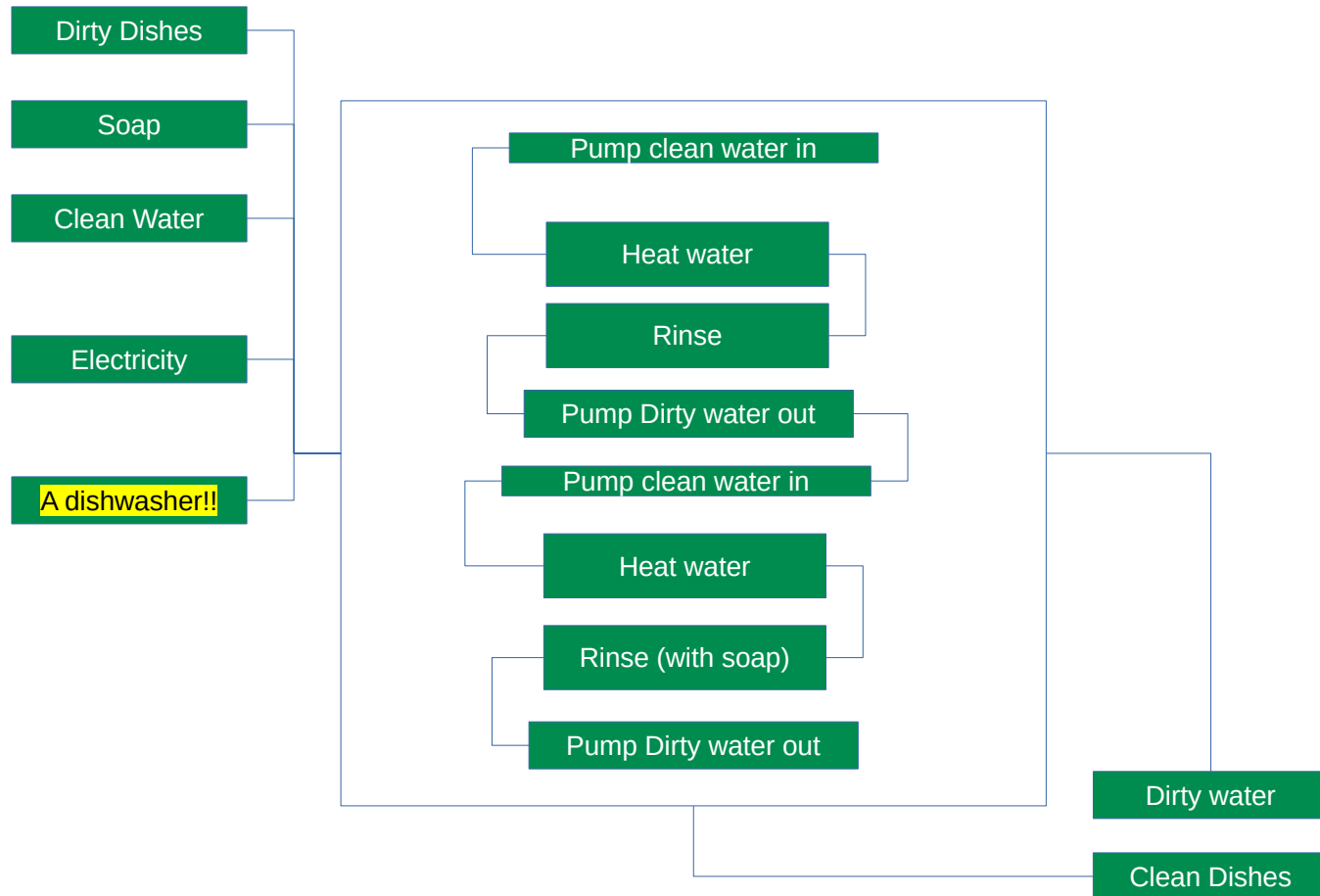
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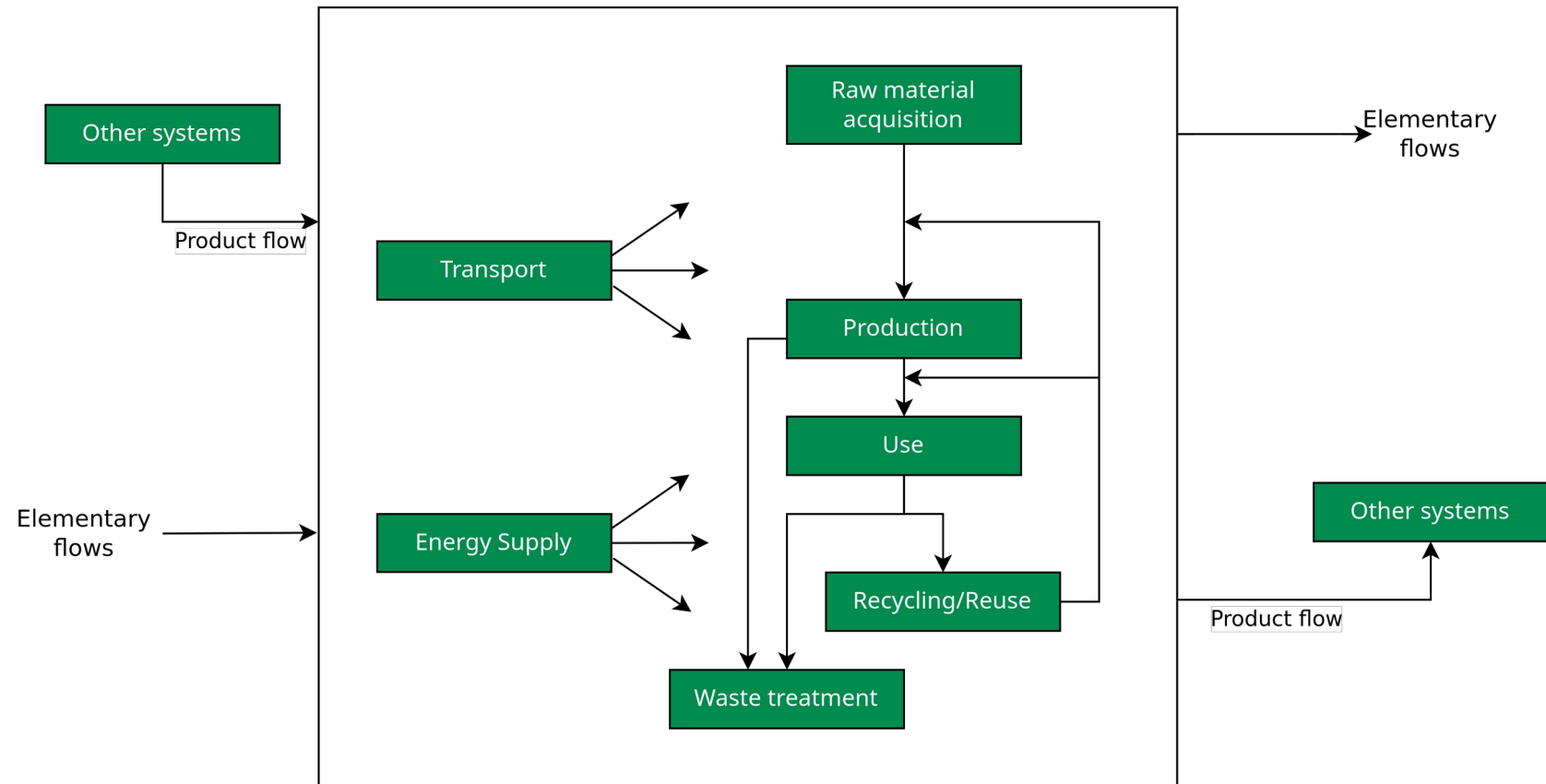
Example of a product system

Lifecycle Inventory Analysis (LCI)

Unit flows for a Dishwasher vs handwash?



Lifecycle Inventory Analysis (LCI)

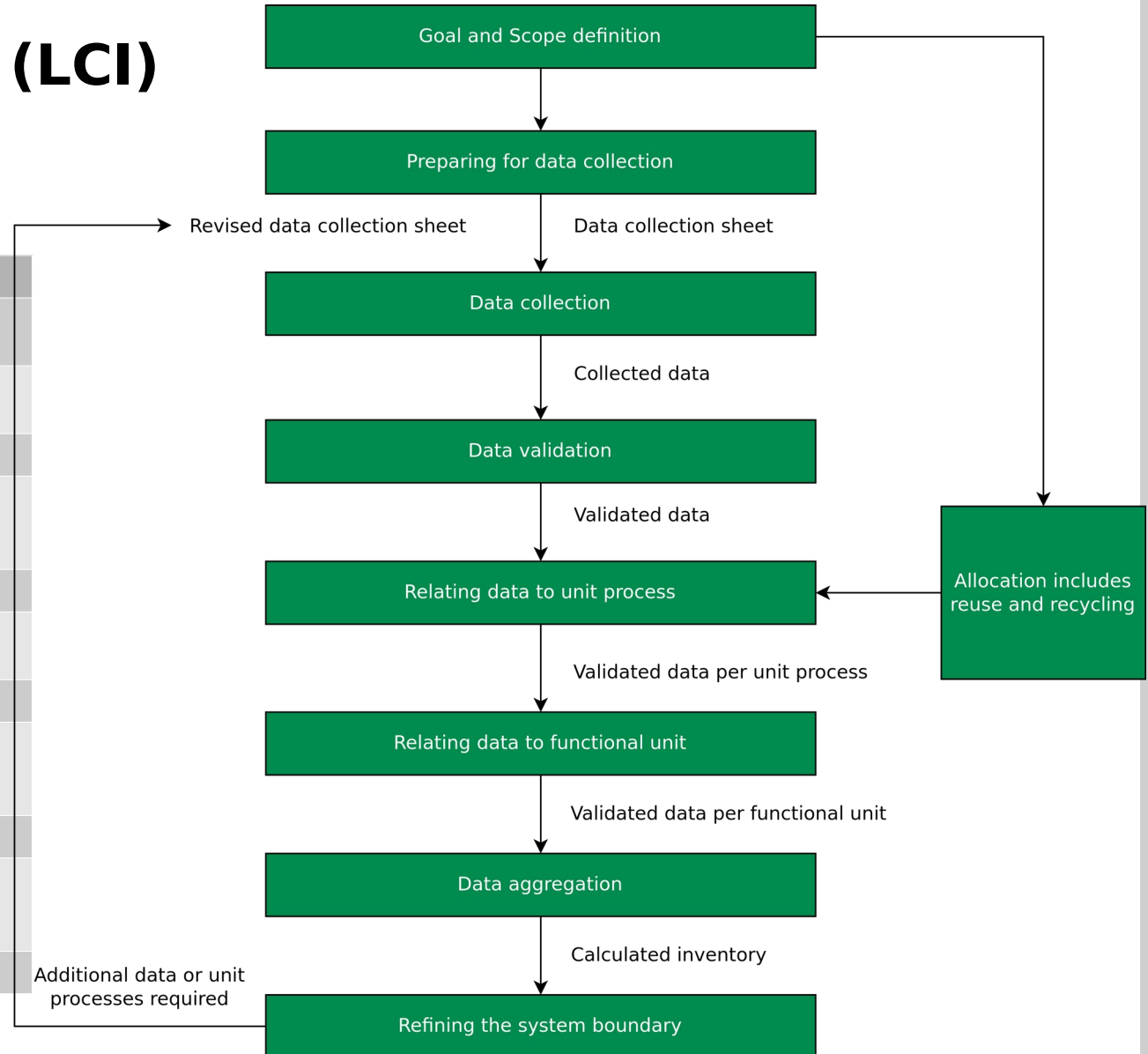


Example of a product system

Lifecycle Inventory Analysis (LCI)

Preparing for data collection

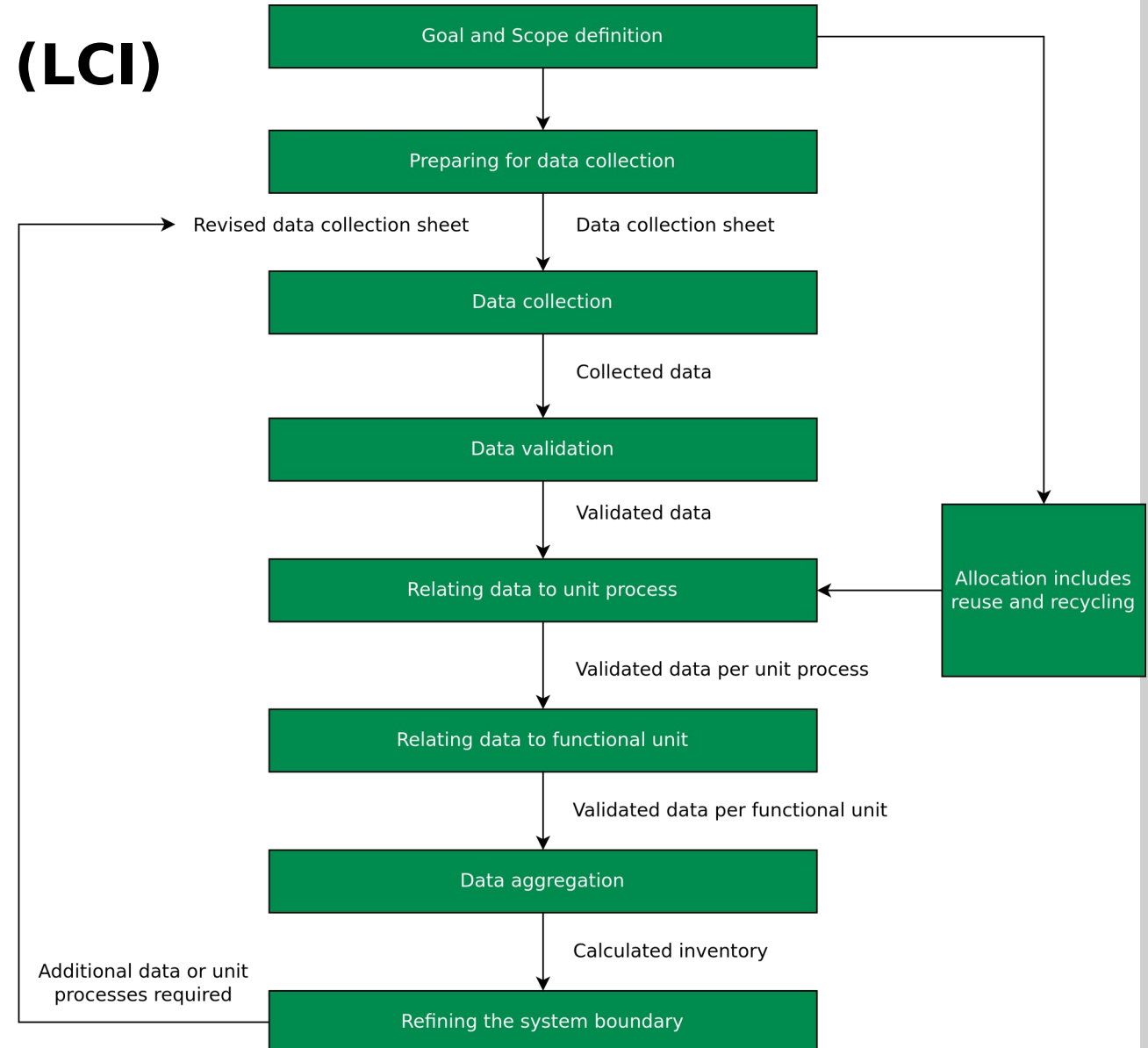
Completed by:	Date of completion:			
Unit process identification:	Reporting location:			
Time period: Year	Starting month:	Ending month:		
Description of unit process:				
Material inputs	Units	Quantity	Description of sampling procedures	Origin
...	...	...	...	...
Water consumption	Units	Quantity		
...	...	...	...	...
Energy Inputs	Units	Quantity	Description of sampling procedures	Origin
...	...	...	...	...
Material outputs	Units	Quantity	Description of sampling procedures	Destination
...	...	...	...	...



Lifecycle Inventory Analysis (LCI)

Data collection and validation

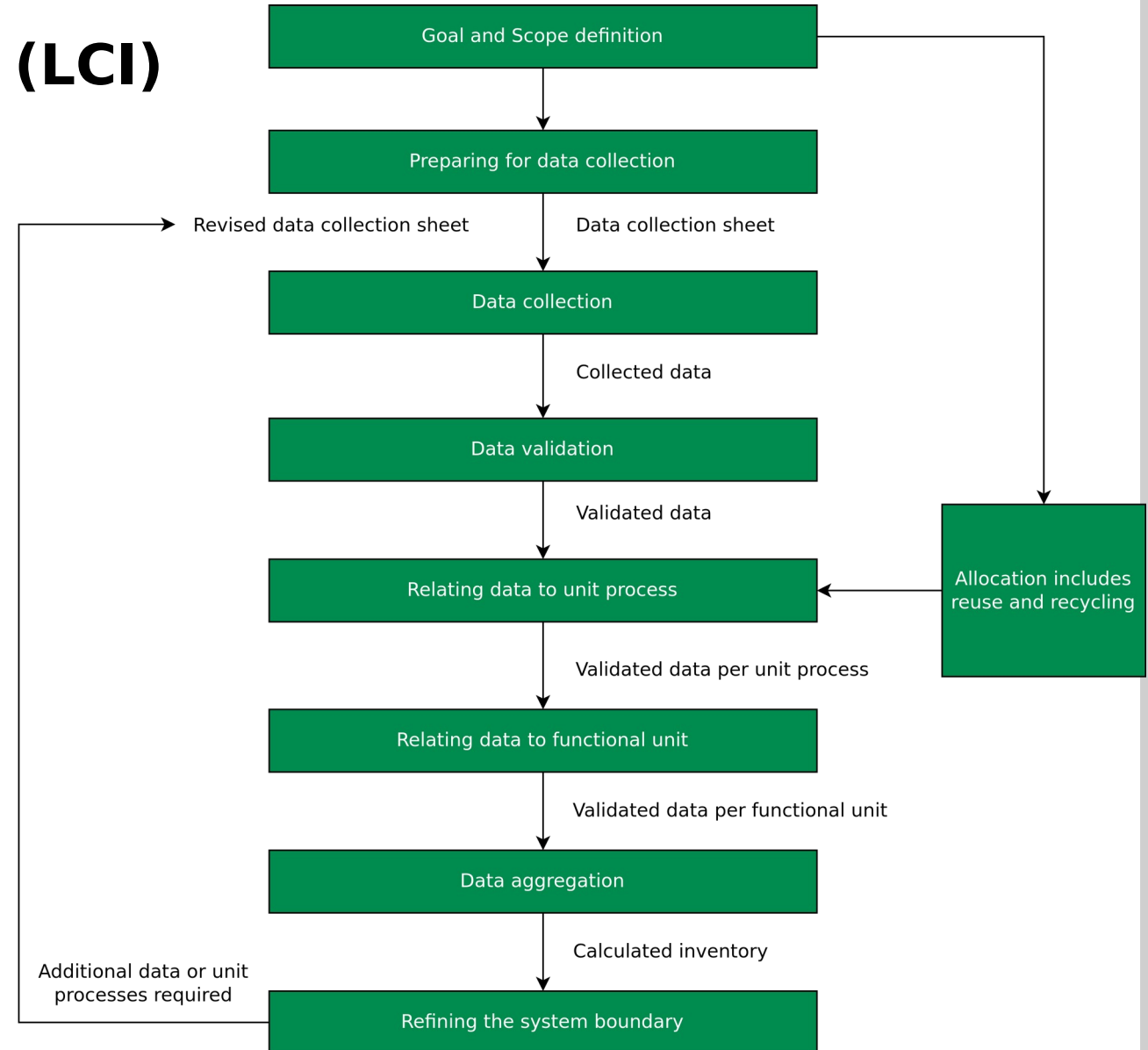
- Data must be validated to confirm and provide evidence for data quality requirements, both during and after the data collection process.
- This can also involve establishing mass and energy balances.
- Obvious anomalies can necessitate collecting alternative data.



Lifecycle Inventory Analysis (LCI)

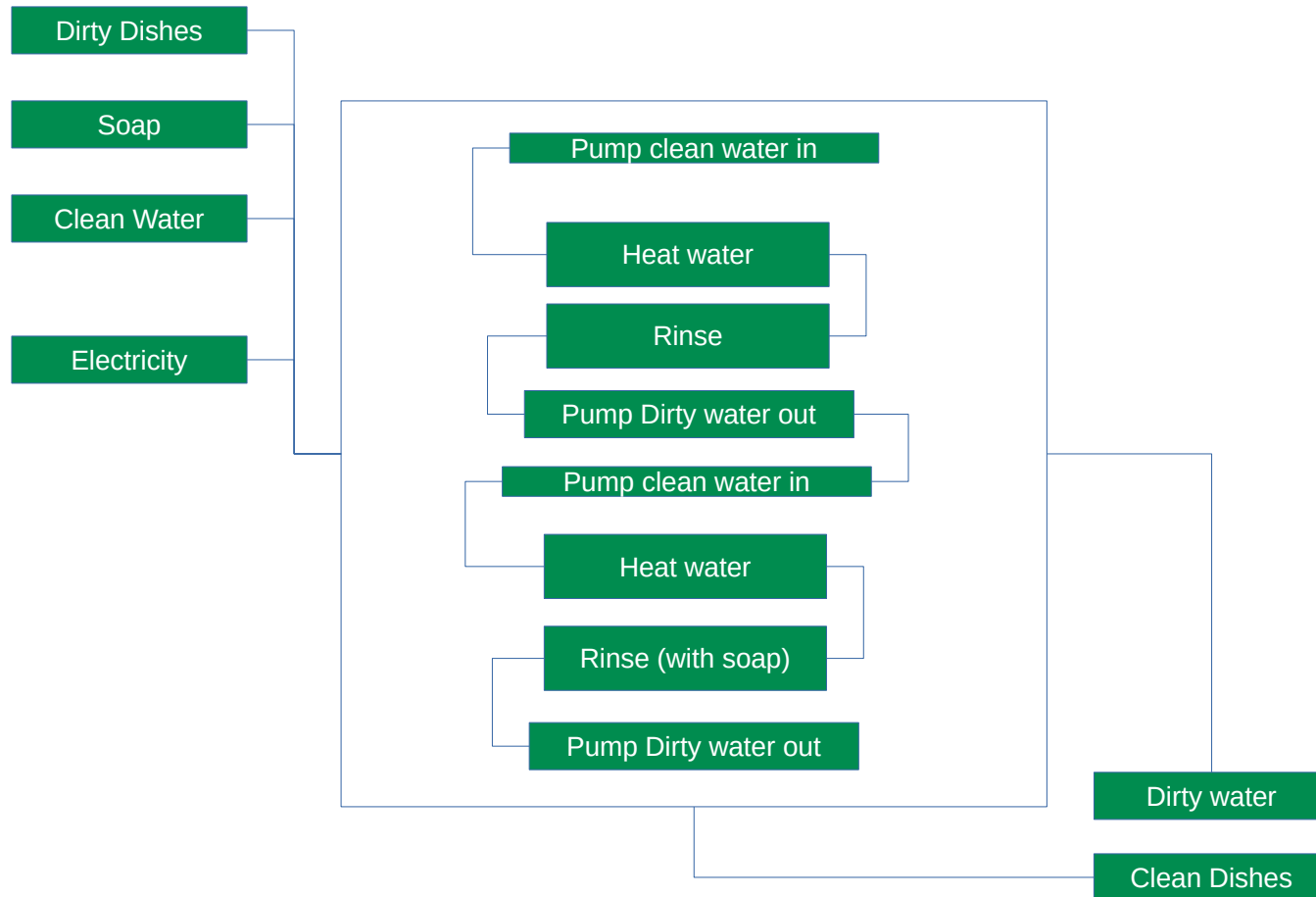
Relating data to unit process and functional unit

- Based on the flow chart and the flows between unit processes, the flows of all unit processes are related to the reference flow.
- The calculation should result in all system input and output data being referenced to the functional unit.
- Recall: a measure of the product(s) or product parts required to deliver the performance defined by the functional unit.



Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit

■ **MushR example**

- Experimental estimation:
 - Colonizable volume of a 5L substrate bag weighing 30g = 3L
 - *Since we have to fold the bag a couple times to seal it.*

Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit

■ MushR example

- Experimental estimation:
 - Colonizable volume of a 5L substrate bag weighing 30g = 3L
 - *Since we have to fold the bag a couple times to seal it.*
 - Colonizable volume of a 3L substrate bucket weighing 90g = 3L (*no change*)

Lifecycle Inventory Analysis (LCI)

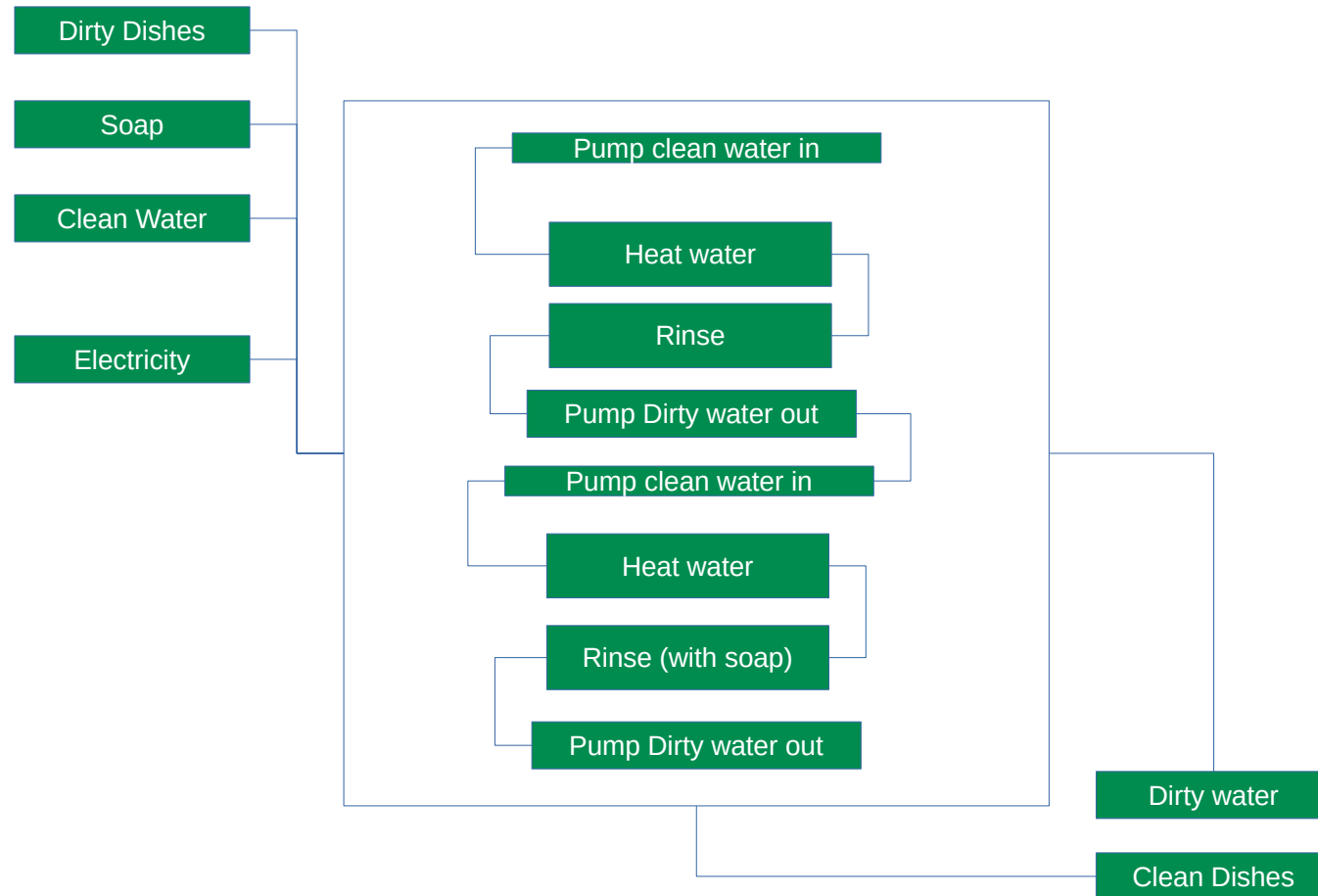
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■ MushR example

- Experimental estimation:
 - Colonizable volume of a 5L substrate bag weighing 30g = 3L
 - *Since we have to fold the bag a couple times to seal it.*
 - Colonizable volume of a 3L substrate bucket weighing 90g = 3L (*no change*)
- This allows us to scale and compare the two container types on similar terms, e.g:
 - 1000g of substrate bags, can contain $(1000 \times 3 / 30) = 100$ Liters of substrate.
 - 1000g of substrate buckets, can contain $(1000 \times 3 / 90) = 33.333$ Liters of substrate.

Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

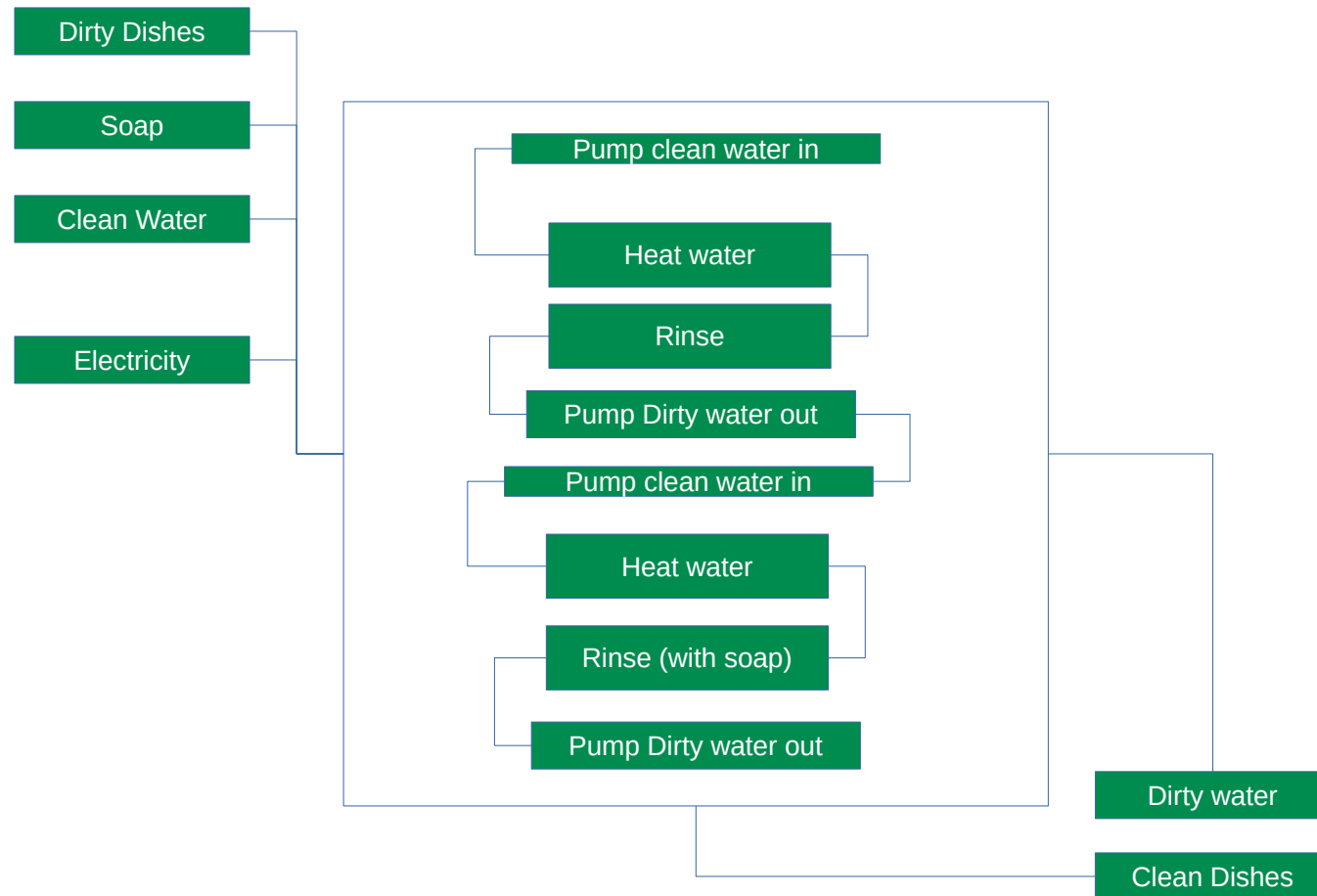
— Experimental estimation:

- 1 wash-cycle with 20 dishes, 20 cups and 20 cutlery items used **1.17 kWh** of energy, **12 Litres** of clean water and 30g of soap.

— But the functional unit was: “2 million dishes, cups, cutlery washed.”

Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

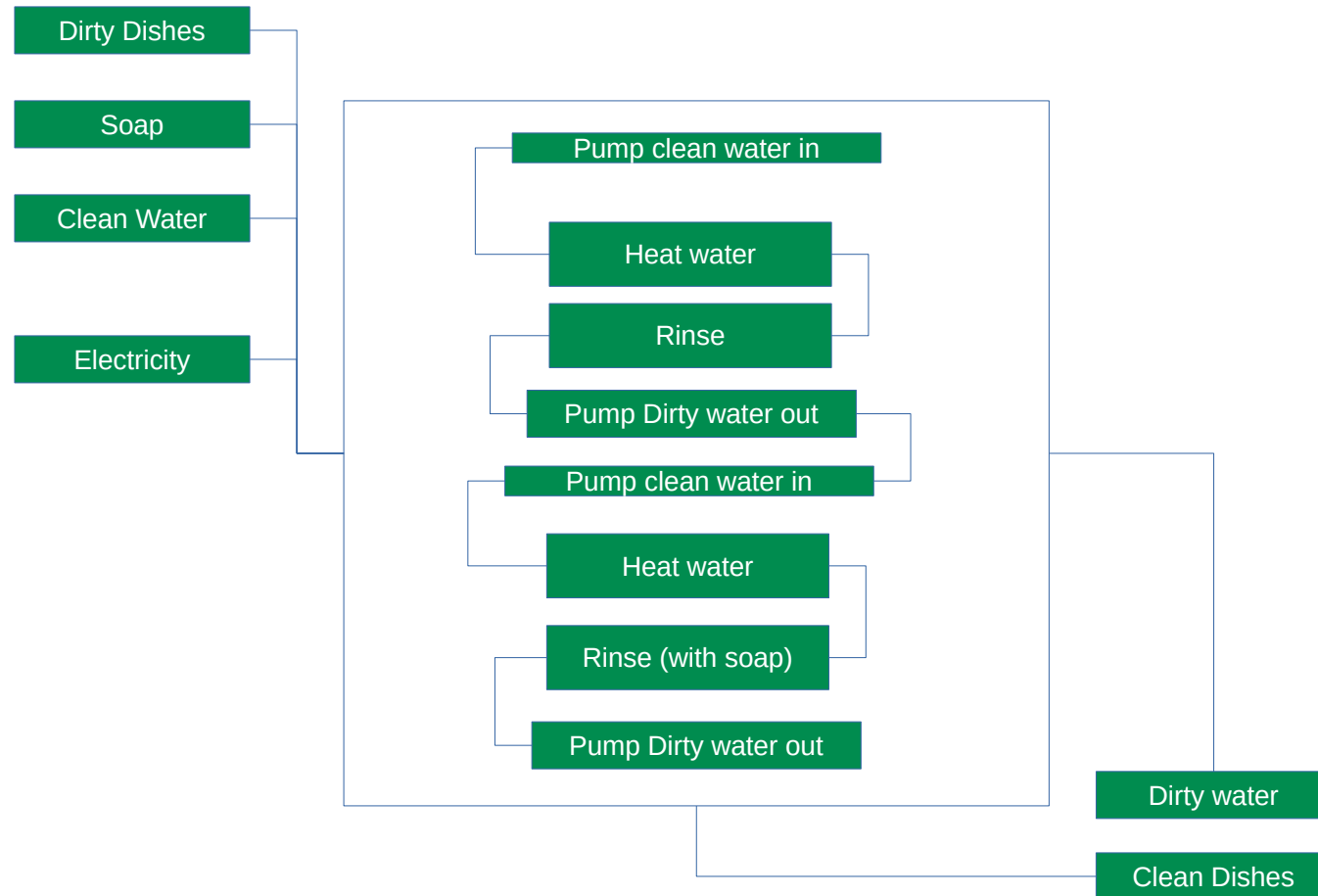
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- But the functional unit was: “2 million dishes, cups, cutlery washed”, so we simply multiply the above numbers by 1,000,000:
1.17 GWh, 12 million litres, 30,000 Kg of soap.

Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

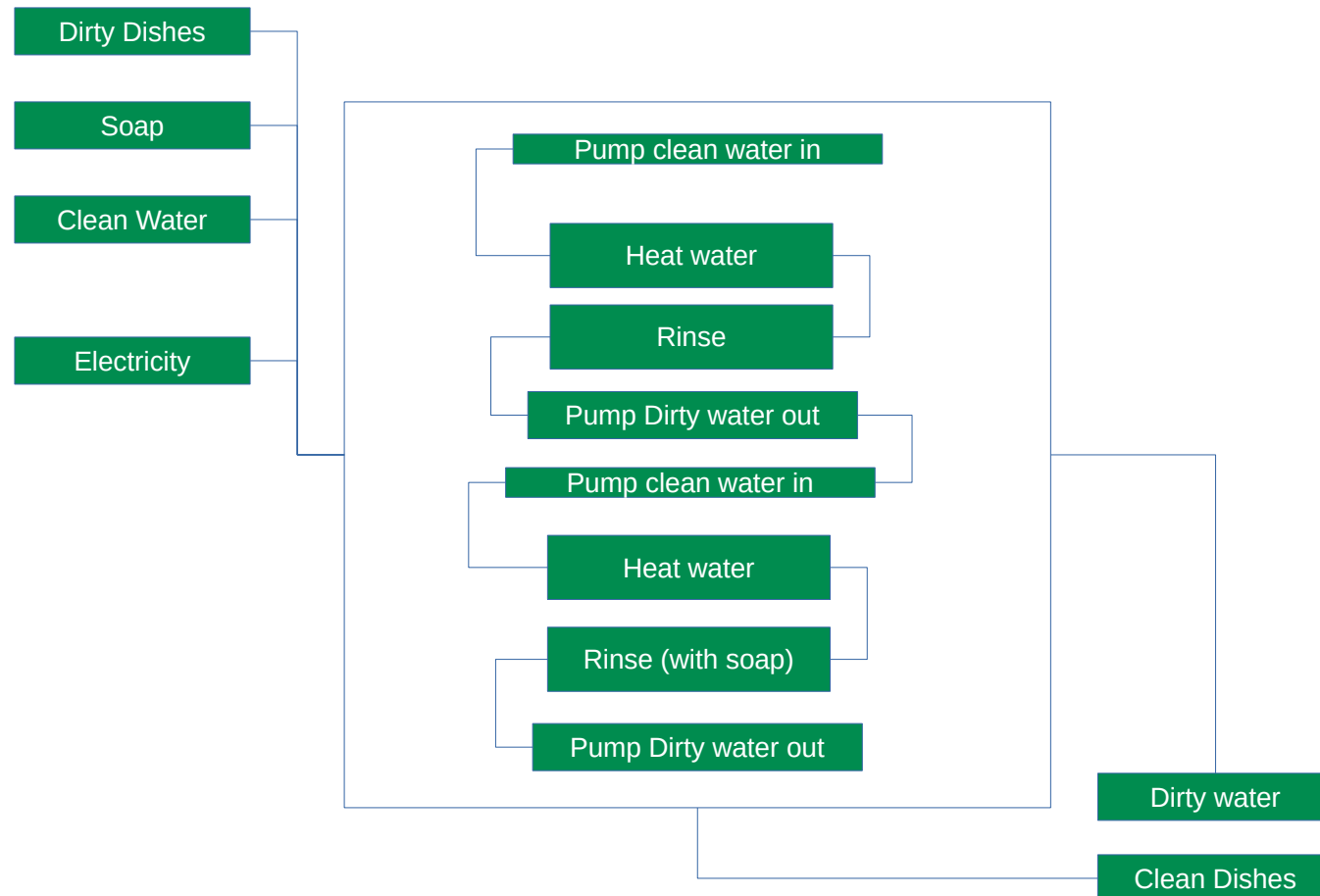
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Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit

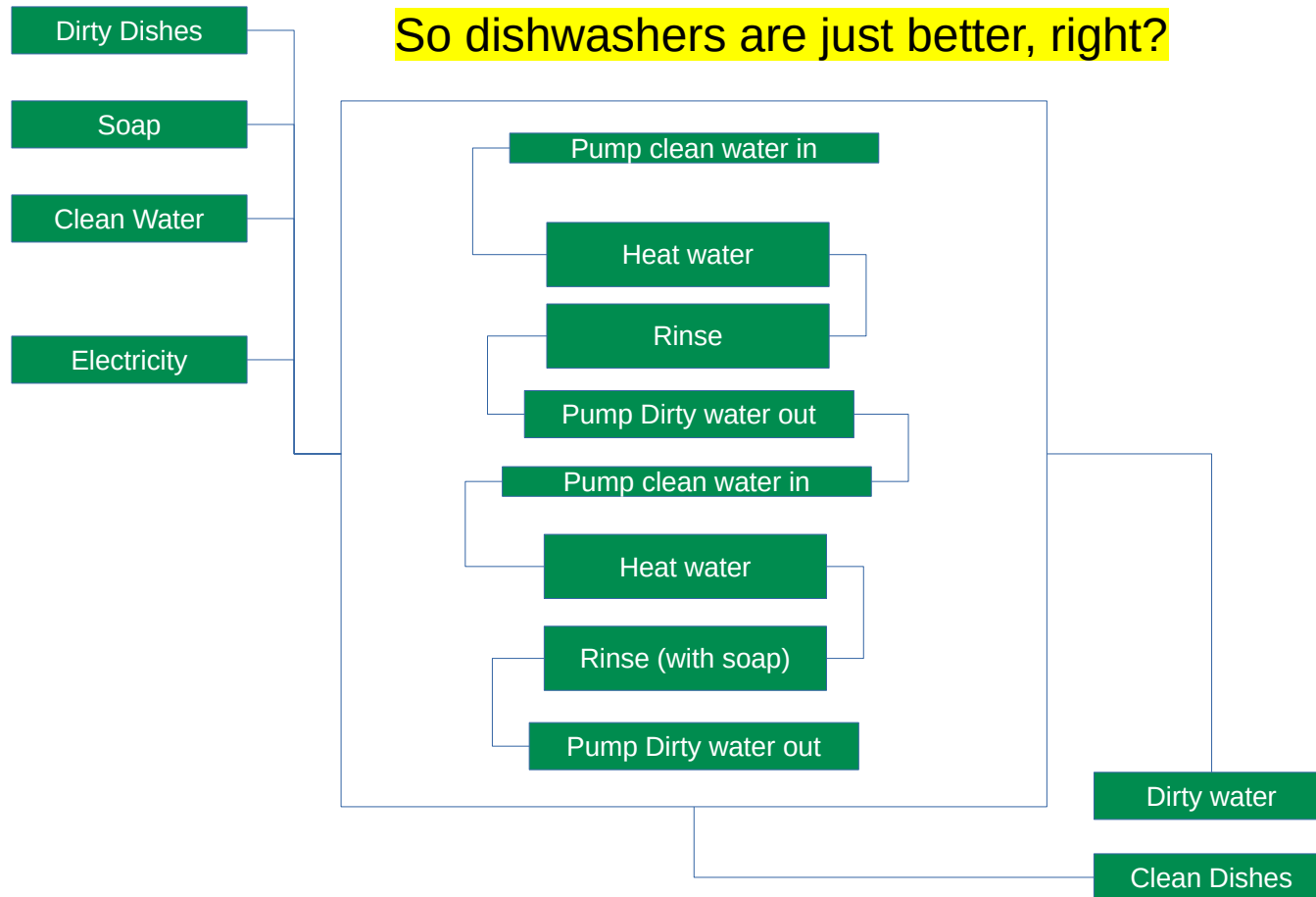


■ Handwash

- Experimental (e.g.) estimation:
 - Washing 2 dishes, cups and cutlery by hand required 0.1 kWh to heat the water, 2 Litre of water, 40g of Soap.
- But the functional unit was:
“2 million dishes, cups, cutlery washed”, so we simply multiply the above numbers by 10,000,000:
10 GWh, 20 million litres, 400,000 Kg of soap.

Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

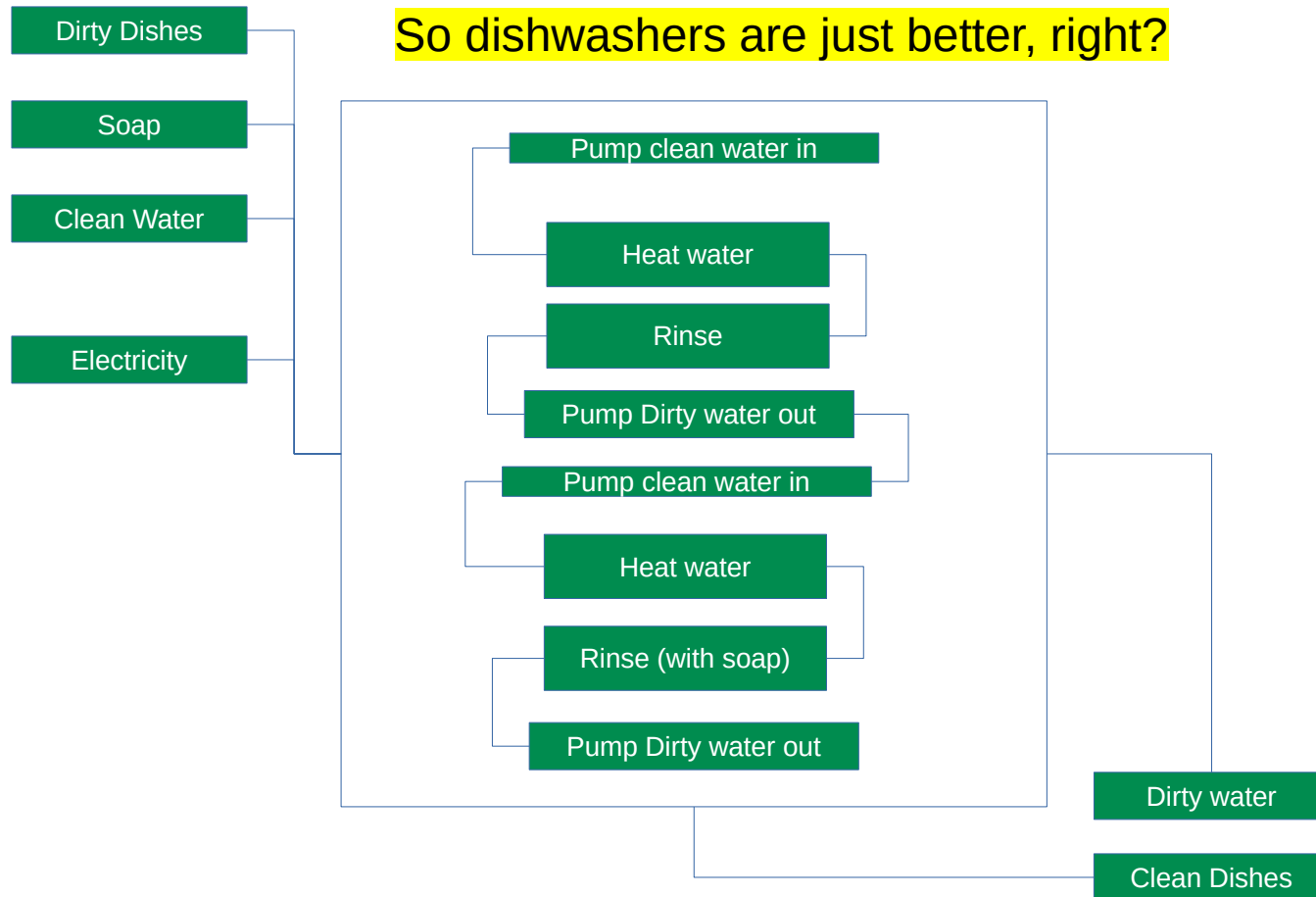
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Lifecycle Inventory Analysis (LCI)

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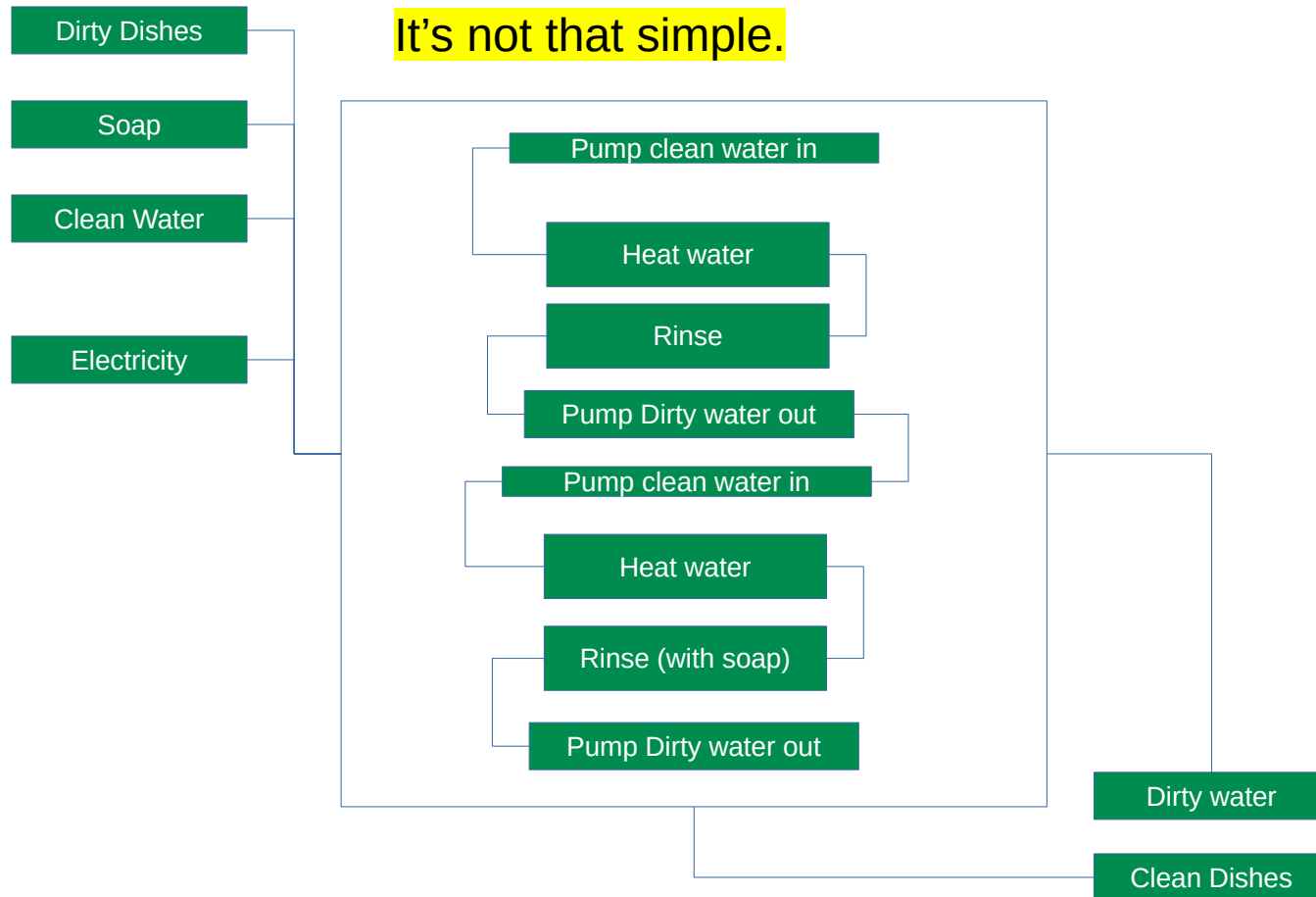
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Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



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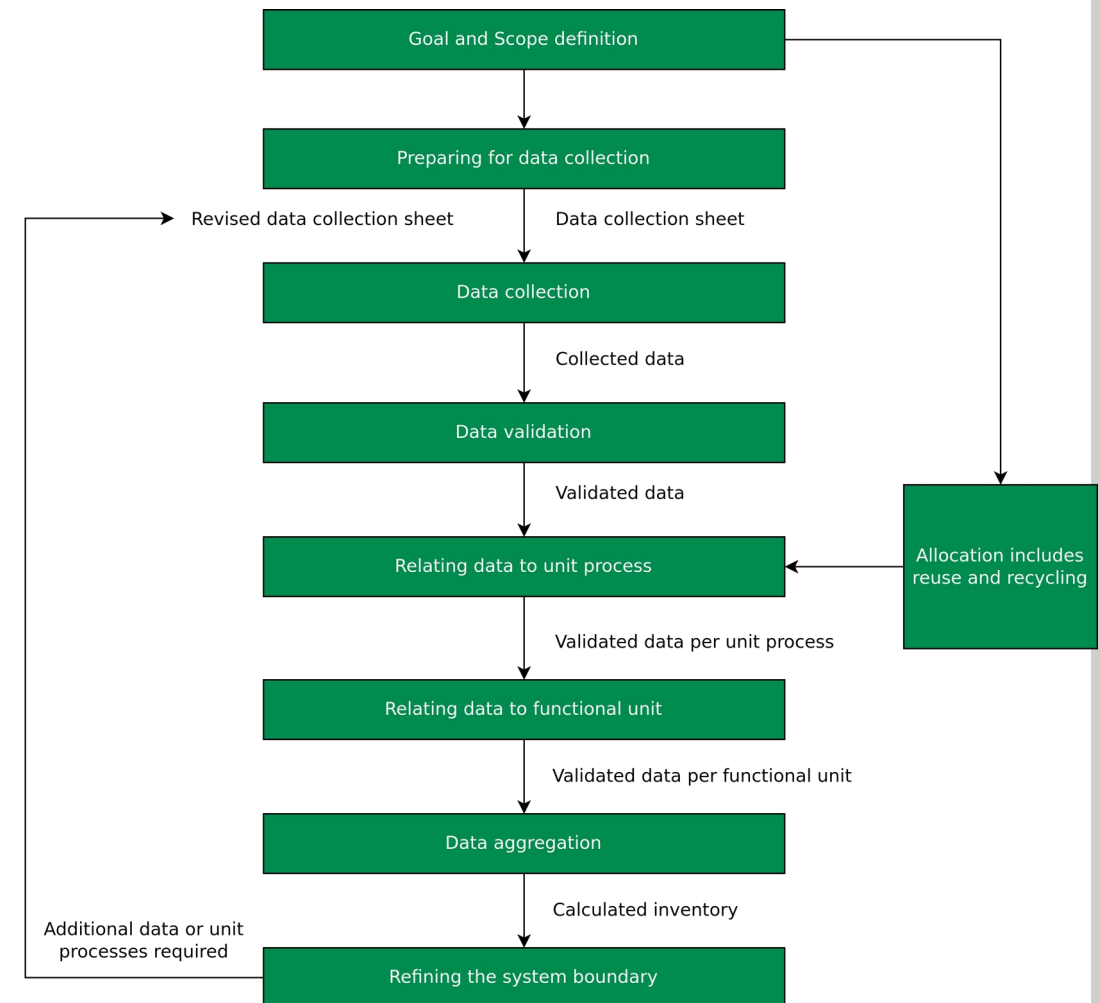
— But the functional unit was: “2 million dishes, cups, cutlery washed”, so we simply multiply the above numbers by **1,000,000**:

1.17 GWh, 12 million litres, 30,000 Kg of soap + **200 dishwashers!**

Lifecycle Inventory Analysis (LCI)

Refining the system boundary

- The initial system boundary is revised, in accordance with the cut-off criteria established before.
- Further analysis may result in:
 - Exclusion of life cycle stages or unit processes if they lack significance
 - Exclusion of inputs or outputs
 - Inclusion of new unit processes, inputs and outputs that are shown to be more significant than estimated before.



Summary

- A deep-dive into Life Cycle Assessment
 - Goal and Scope definition
 - Functional Units & Reference Flows
 - Life Cycle Inventory analysis
 - Unit Processes and Process flows
 - Relating LCI data to functional units
- Examples from Polestar, 2020 EU Commission report, MushR project, Dishwasher vs Handwashing

EXERCISE 3 – LCA – PART 1

Exercise 3 - LCA - Part 1

- **Main Goal:** Compare the water+power consumption of washing machines.
- **Main Task:** Defining Goal + Scope, Define unit processes, and determine all the unit input and output flows of the system. → Interactive Exercise session (Now)
- **Homework:** Collect LCI data for Unit Processes and relate it to the functional units. (Refer to Exercise Sheet on GitHub)

Questions?